Worksheet 4

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1. 4 candidates are running for mayor; Adams, Brown, Collins and Dalton. The following probabilities are assigned:

   $P[$ Adams wins $] = 0.42$ $\quad P[$ Brown wins $] = 0.09$

   $P[$ Collins wins $] = 0.27$ $\quad P[$ Dalton wins $] = 0.22$

   $0.42 + 0.27 + 0.09 + 0.22 = 1$

   Determine the probabilities for the following events (use 2 decimal places):

   a. Adams loses. (Adam not win)

   b. Either Brown or Dalton wins.

   c. Adams, Brown, or Collins wins.

(a) $P(\text{Adam lose}) = P((\text{Adam wins})^c)$

$= 1 - P(\text{adams wins})$

$= 1 - 0.42 = 0.58$

$= 0.27 + 0.09 + 0.22$

(b) $P(\text{Brown wins} \overset{\cup}{\text{or}} \text{Dalton wins})$

$= P(\text{Brown wins}) + P(\text{Dalton wins})$

$= 0.09 + 0.22 = 0.31$

(c) $P(\text{adams win} \cup \text{Brown win} \cup \text{Collins wins})$

$= P(\text{adams win}) + P(\text{Brown wins}) + P(\text{Collins win})$

$= 0.09 + 0.27 + 0.42 = 0.78$

$P(A \overset{\wedge}{\text{and}} B) = 0$

2. $P[A] = 0.30$ and $P[B] = 0.40$. If A and B are mutually exclusive events, what is $P[A \cup B]$?

3. $P[A] = 0.60$ and $P[B] = 0.70$. If A and B are independent events, what is $P[A \cup B]$?

(2) $P(A \cup B) \overset{\text{addition Rule}}{=} P(A) + P(B) - \underbrace{\cancel{P(A \cap B)}}_{0}$

$= P(A) + P(B)$

$= 0.3 + 0.4 = 0.7$

mutually Ex :
$P(A \cap B) = 0$

(3) $P(A \cap B) \overset{\text{indep}}{=} P(A) \cdot P(B)$

$P(A \cup B) \overset{\text{addition Rule}}{=} P(A) + P(B) - P(A \cap B)$ by def of indep.

$= P(A) + P(B) - P(A) \cdot P(B)$

$= 0.6 + 0.7 - 0.6 \cdot 0.7$

$= 0.88$

4. Assume you have applied for two scholarships, a Merit scholarship (M) and an Athletic scholarship (A). The probability that you receive an Athletic scholarship is 0.18. The probability of receiving both scholarships is 0.11. The probability of getting at least one of the scholarships is 0.3.

   a. What is the probability that you will receive a Merit scholarship?

   b. Are events A and M mutually exclusive? Why or why not? Explain.
   No. Since $P(A \cap M) = 0.11 \neq 0$

   c. Are the two events A, and M, independent? Explain, using probabilities.
   $P(A) = 0.18$ $P(M) = 0.23 \rightarrow P(A) \times P(M) = 0.0414 \neq 0.11 = P(M \cap A) \rightarrow$ No

   d. What is the probability of receiving the Athletic scholarship given that you have been awarded the Merit scholarship?

   e. What is the probability of receiving the Merit scholarship given that you have been awarded the Athletic scholarship?

$P(A \cap B) = P(A) \times P(B)$

$P(A|B) = P(A)$ or $P(B|A) = P(B)$

$P(A) = 0.18$ $P(A \overset{\cap}{\text{and}} M) = 0.11$

$P(A \overset{\cup}{\text{or}} M) = 0.3$

(a) $P(M) = ?$

$P(A \cup M) = P(A) + P(M) - P(A \cap M)$

$0.3 = 0.18 + P(M) - 0.11$

$-0.18 + 0.11$

$\rightarrow P(M) = 0.41 - 0.18 = 0.23$

(d) $P(A|M) \overset{\text{def Condi Prob}}{=} \dfrac{P(A \cap M)}{P(M)} = \dfrac{0.11}{0.23} = 0.478$

(e) $P(M|A) \overset{\text{def Condi Prob}}{=} \dfrac{P(A \cap M)}{P(A)} = \dfrac{0.11}{0.18} = 0.61$